

Chapter 43. Ventilator-Induced Diaphragmatic Dysfunction

Theodoros Vassilakopoulos

Ventilator-Induced Diaphragmatic Dysfunction: Introduction

Controlled mechanical ventilation (CMV) is a ventilator mode where the respiratory muscles are not contracting, and the ventilator takes full responsibility for inflating the respiratory system. Evidence accumulating over the past two decades has shown that CMV can induce dysfunction of the diaphragm, mainly consisting of atrophy, oxidative stress, and ultrastructural injury, resulting in decreased diaphragmatic contractility. This is called *ventilator-induced diaphragmatic dysfunction (VIDD)*.¹

The frequency with which CMV is used cannot be determined with certainty. An international survey revealed that 13% of mechanically ventilated patients receive a neuromuscular blocker for 8% of the total days of ventilator support.² In these patients, full ventilator support is mandatory. Moreover, it was recently shown that the use of neuromuscular blockers during the first 2 days of ventilator support improves the survival of patients with severe acute respiratory distress syndrome,³ a finding that may lead to increased use of CMV. Other patient groups not receiving neuromuscular blockers also receive full ventilator support, such as patients with traumatic brain injury, postoperative neurosurgical patients, comatose patients, patients with status epilepticus on barbiturate coma to suppress seizure activity, and so on. Thus, a considerable percentage of ventilated patients are at risk of developing VIDD.

The mechanisms of this dysfunction and the clinical relevance for mechanically ventilated patients is the subject of this chapter.

Evidence for Ventilator-Induced Diaphragm Dysfunction in Humans

The first human evidence for the existence of VIDD came from retrospective analysis of postmortem data obtained in neonates who received ventilatory assistance for 12 days or more immediately before death. This study revealed diffuse diaphragmatic myofiber atrophy (small myofibers with rounded outlines), not present in extradiaphragmatic muscles or diaphragms of infants ventilated for 7 days or less.⁴

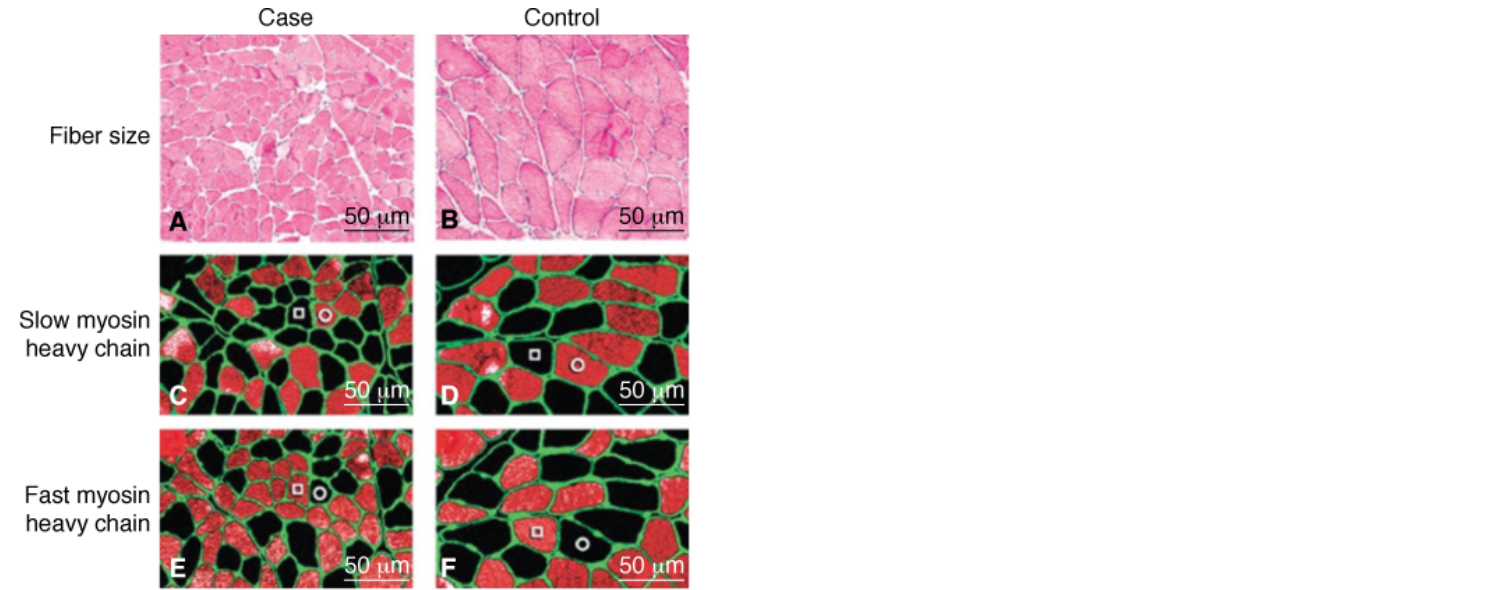
Recently, direct evidence came from studies of brain-dead organ donors (with an intact circulation) who underwent CMV and who developed biopsy-proven atrophy^{5,6} and decline in the contractility of their diaphragm.⁶

The Components of Ventilator-Induced Diaphragm Dysfunction

Atrophy

Ventilator-induced atrophy⁷ is usually measured by the reduction of the cross-sectional area of myocytes in histologic sections. Atrophy has been observed in both slow-twitch and fast-twitch human diaphragmatic fibers and is quite significant (Fig. 43-1), its magnitude amounting to 40% to 50% after quite variable periods of CMV (range: 15 to 276 hours.)^{5,6,8} Atrophy preferentially affects the diaphragm because it was not observed in the pectoralis major muscle of the same patients.⁵ The longer the duration of CMV, the greater the observed atrophy (Fig. 43-2).⁶

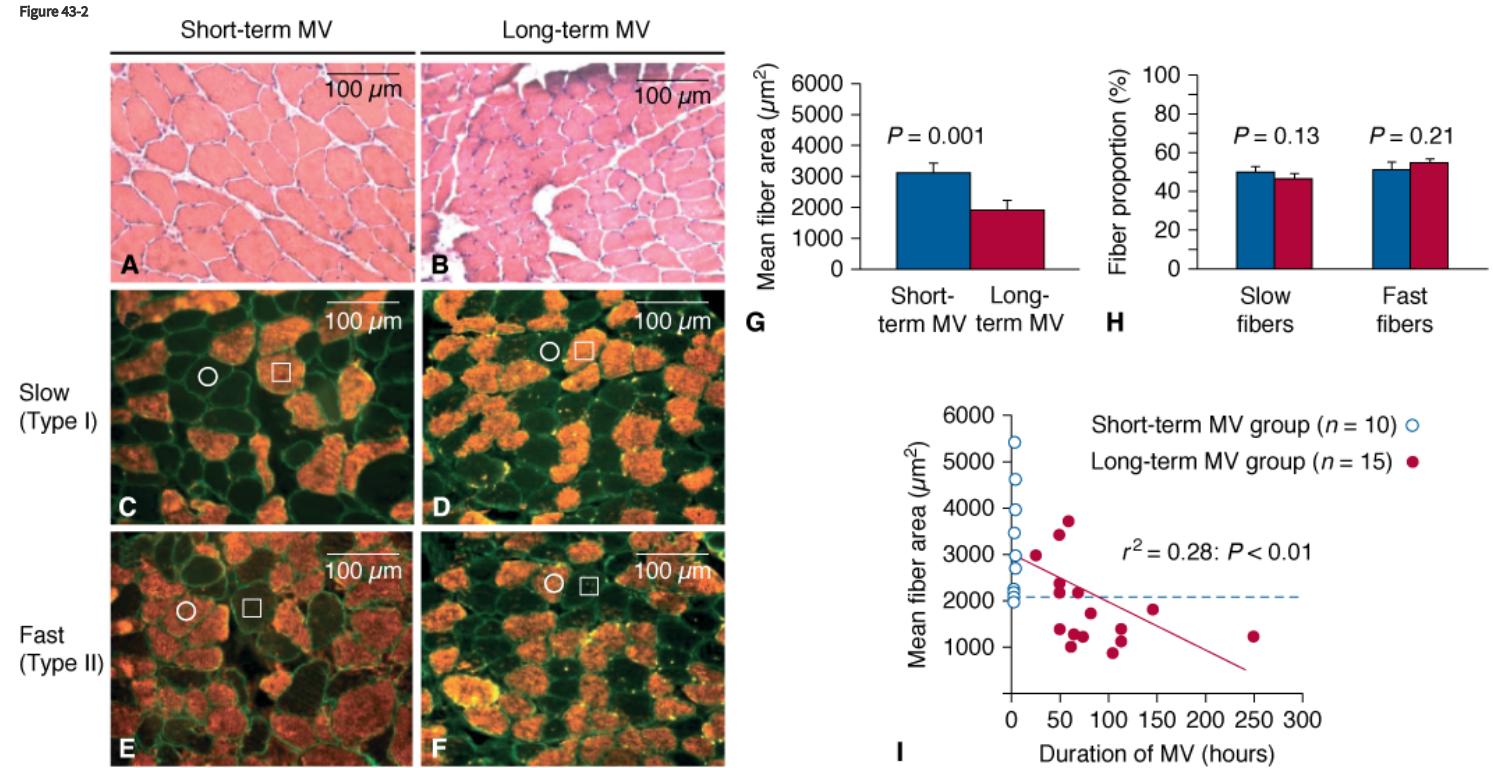
Figure 43-1



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Atrophy in the human diaphragm after CMV. The slow-twitch and fast-twitch fibers in the case specimens (Panels A, C, and E) are smaller than those in the control diaphragms (Panels B, D, and F). Panels A and B (hematoxylin and eosin) show that neither inflammatory infiltrate nor necrosis is present in case or control specimens. The sections in Panels C and D were preincubated with NOQ7.5.4D antibody, which is specific for the slow myosin heavy chain, whereas sections in Panels E and F were preincubated with the MY-32 antibody, which reacts with all fast myosin heavy chains. In addition, in each section, all fibers are outlined by an antibody reactive to laminin. In each of the sections, fibers reacting with the antibody appear orange-red, whereas fibers not reacting with the antibody appear black. In Panels C, D, E, and F, a representative slow-twitch fiber is indicated by an open circle and a fast-twitch fiber by an open square.

(Used, with permission, from Levine et al.⁵)



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Relationship between duration of mechanical ventilation (MV) and diaphragmatic atrophy. Representative images of transverse frozen sections obtained from the diaphragms of the short-term MV (A,C,E) and long-term MV (B,D,F) groups are shown. The diaphragmatic sections are stained with hematoxylin and eosin (A, B) or with antibodies directed against slow (C, D) or fast (E, F) isoforms of myosin heavy chain. C to F. Serial sections, individual representative slow-twitch and fast-twitch fibers are marked by an open square and circle, respectively. G and H. Quantitative analyses of diaphragmatic fiber size (mean cross-sectional

area) and fiber-type proportions in the two patient groups, respectively. I. The significant correlation between the degree of diaphragmatic atrophy and the duration of MV (*horizontal dashed line* indicates lowest value of fiber size measured in short-term MV group). (Used, with permission, from Jaber et al.⁶)

Animal studies also suggest that atrophy is more pronounced in the diaphragm, which atrophies earlier than the peripheral skeletal muscles, which are also inactive during CMV.^{9–11} Two days of CMV with positive end-expiratory pressure (PEEP) (2 cm H₂O) induced atrophy in rabbits,¹² whereas 3 days of CMV without PEEP were inadequate to induce atrophy¹³ in the same species, which suggests that the rapidity of atrophy development might be augmented with use of PEEP. The increased lung volume at the end of expiration (with use of PEEP) places the passive diaphragm in a relatively shortened position and skeletal muscle atrophies faster when shortened.^{14,15}

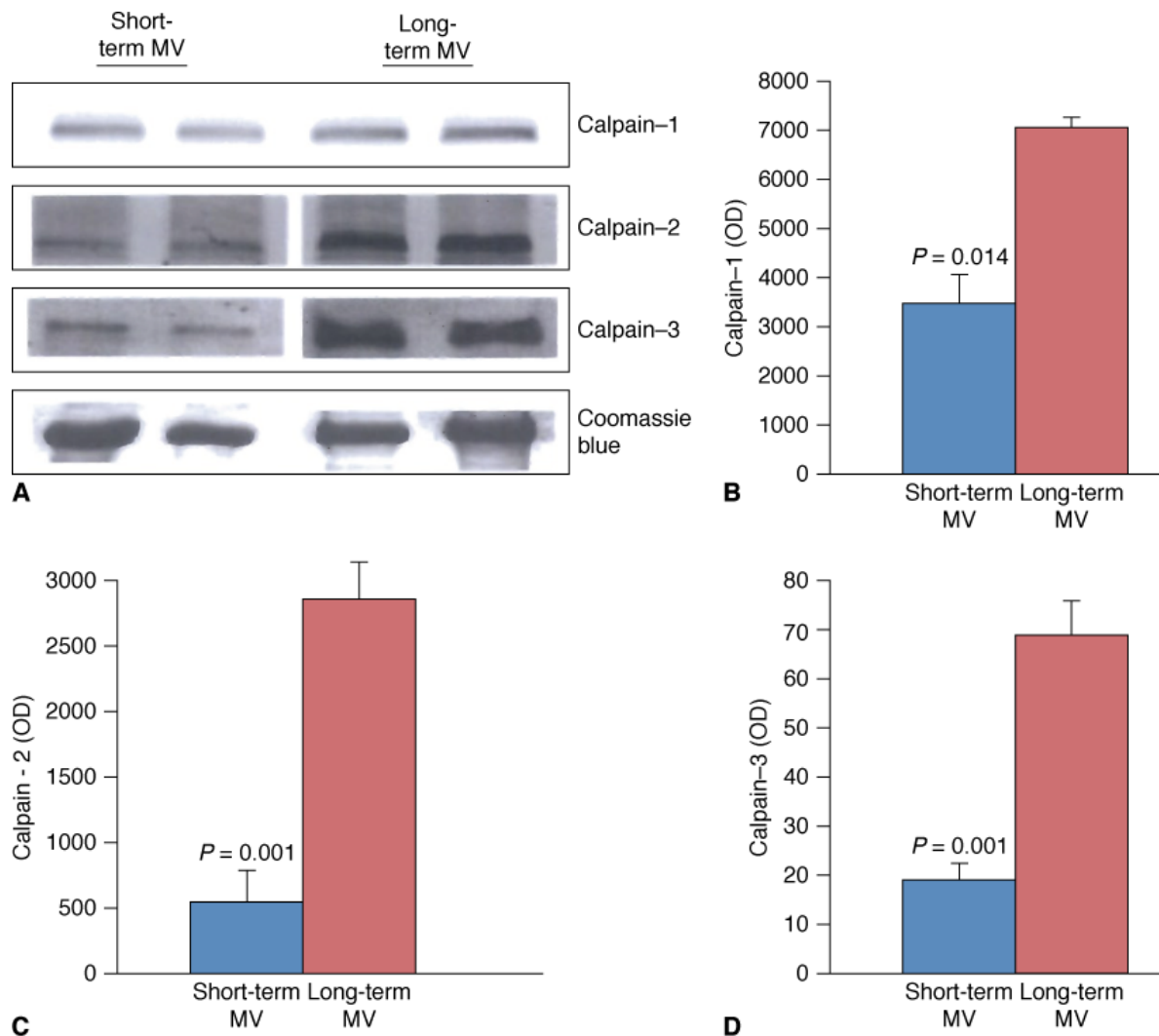
Atrophy can result from decreased protein synthesis, increased proteolysis, or both. Six hours of CMV in rats decreased the in vivo rate of mixed muscle protein synthesis (averaged for all muscle proteins) by 30% and the rate of myosin heavy-chain protein synthesis by 65%, both of which persisted throughout 18 hours of CMV.¹⁶ In addition, 24 hours of CMV suppressed the messenger RNA levels of insulin-like growth factor-1, which stimulates protein synthesis.¹⁷ Thus, CMV decreases protein synthesis in the diaphragm.

Increased proteolysis has been documented in diaphragmatic strips of animals subjected to 18 hours of CMV.⁹ Mammalian cells have four different protein systems and organelles for proteolysis—the calpains,⁶ the caspases,⁵ the proteasome system,^{5,8} and the autophagy-lysosomal system⁸—all of which are activated in the human diaphragm after CMV, a finding previously suggested by animal models.^{9,18–20}

Calpains

Calpains, which are activated after CMV (Fig. 43-3), do not fully degrade, but only partially cleave proteins in vivo. This renders the proteins amenable to the proteasome.⁷ The stimulus for calpain activation is not known, but calcium elevation in the cell is a prerequisite. The reduced (messenger RNA) levels of sarcoplasmic reticulum calcium adenosine triphosphatase (ATPase; the enzyme that removes calcium from the sarcoplasm), secondary to 24 hours of CMV²¹ in animal models may contribute to calpain activation.

Figure 43-3



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Expression of calpain isoforms in diaphragms of short-term and long-term mechanical ventilation (MV) groups. Representative immunoblots (A) and group mean quantification of protein levels measured in diaphragmatic tissues obtained from the short-term and long-term MV groups for calpain-1 (B), calpain-2 (C), and calpain-3 (D). (Used, with permission, from Jaber et al.⁶)

Caspases

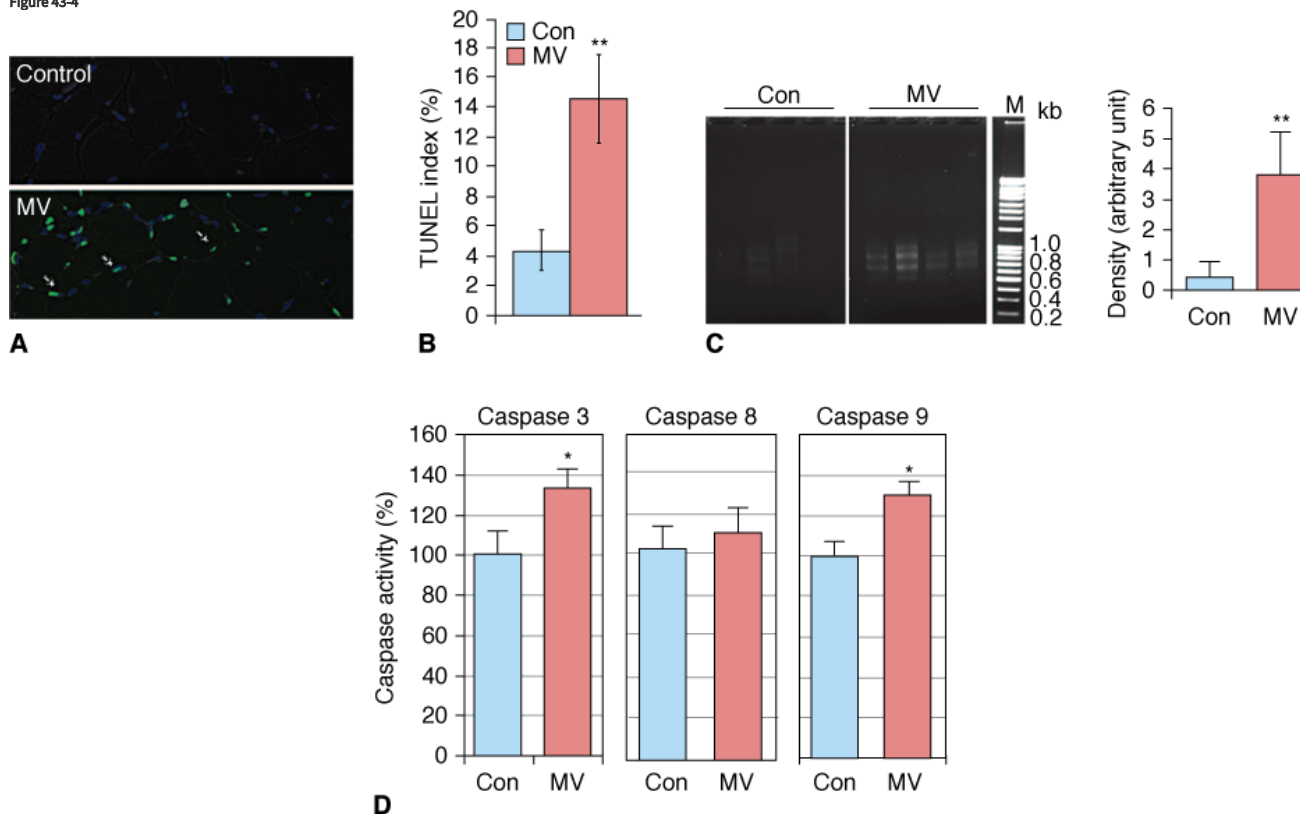
Caspases are proteases that can degrade proteins and especially complexes of actin and myosin^{18,22} and release them from the myofibrillar lattice.

Upregulation of caspase-3 expression has been documented in the diaphragm secondary to CMV in both humans⁵ and in animal models of VIDD.¹⁹ Caspase-3 can be activated by oxidative stress, increased intracellular calcium, and increased calpain activity.¹⁸

In animal models of VIDD, the decreased volume of the cytoplasm (atrophy) was observed in the presence of decreased number of myonuclei (skeletal muscle cells are multinucleated cells and theoretically a single myonucleus can sustain the necessary gene expression for a limited area of the cytoplasm, a relationship known as the myonuclear domain²³), so that the myonuclear domain remains constant.¹⁹ This decrease in myonuclear content was mediated by a caspase-3 dependent increase in apoptosis, which was evident as early as 6 hours after the onset of CMV.¹⁹ Both the apoptosis and atrophy were attenuated with caspase-3 inhibition.¹⁹

CMV-induced apoptosis has also been documented in the human diaphragm (Fig. 43-4).²⁴ The finding of elevated caspase 9, but not caspase 8, and elevated Bcl2-interacting mediator of cell death (Bim) (and its transcriptional variants), but not Fas or Fas Ligand, suggest that the intrinsic (mitochondrial), rather than the extrinsic apoptotic pathway, is the primary pathway that operates in CMV-induced apoptosis in the human diaphragm, resulting in caspase 3 activation and nuclear DNA fragmentation.²⁴ Complementary experiments, conducted in vitro, suggest that oxidative stress is a potential proximal activator of *all* of the pathways that have been implicated in VIDD, with mitochondrial dysfunction representing a central component of this cascade.²⁴ Bcl2-interacting mediator of cell death is probably an important mediator of the oxidative stress-induced activation of the intrinsic apoptotic pathway, and Fos, FoxO1, and Stat3 are the transcription factors that regulate its expression.²⁴

Figure 43-4



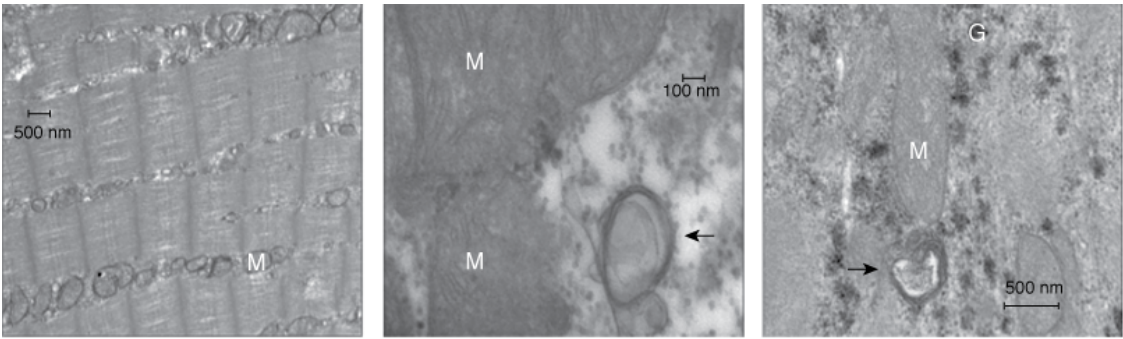
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Apoptosis in the human diaphragm after CMV. CMV results in DNA fragmentation and activates caspases 3 and 9 in human diaphragm. **A**. TUNEL (TdT [terminal deoxynucleotidyl-transferase]-mediated [2'-deoxyuridine 5'-triphosphate]-biotin nick-end labeling) staining was performed on cryosections of control and ventilated human diaphragm. Fragmented genomic DNA was labeled with FITC (fluorescein isothiocyanate) conjugated dUTP (deoxyuridine triphosphate); positive signals appear green. Myonuclei are visualized by DAPI (4,6-diamidino-2-phenylindole) staining (blue). Note that the positive TUNEL staining signals (green) are localized in nuclei stained by DAPI (blue). **B**. TUNEL-positive nuclei were counted and normalized to total myonuclei; ratio is shown as the TUNEL index. Control (Con), $n = 9$; MV, $n = 10$. **C**. DNA fragmentation was measured by polymerase chain reaction (PCR)-based detection and visualized by electrophoresis in 2% agarose gel. Density of the PCR products was quantitated with Image J and normalized to the total DNA input. M, DNA markers. Control, $n = 5$; MV, $n = 8$. **D**. Caspase 3, 8 and 9 enzymatic activities from control and MV human diaphragm lysates were measured by fluorometric assay. Results are presented as relative fluorescence units after normalization to total protein amount. Control, $n = 7$; MV, $n = 9$. *, $P < 0.05$; **, $P < 0.01$. (Used, with permission, from Tang et al.²⁴)

Autophagy–Lysosome Pathway

The autophagy–lysosome pathway is also upregulated in the ventilated human diaphragm.⁸ Autophagy (Greek for self-eating) is a catabolic pathway characterized by the formation of vesicles (autophagosomes) that engulf cytoplasmic organelles and proteins, which then fuse with lysosomes that degrade their contents. This process is a major mechanism for degrading long-lived proteins and organelles. Autophagy occurs at low basal levels to perform homeostatic functions, but can be rapidly upregulated when cells need to generate energy. Hussain et al.⁸ demonstrated that prolonged CMV triggers the appearance of autophagosomes (Fig. 43-5) and increases the expression of autophagy-related genes in the diaphragm (Fig. 43-6). This is associated with upregulation of the transcription factor FOXO-1. It should be emphasized that these changes are relatively specific to the diaphragm and are not observed to the same degree in the control limb muscle (quadriceps) of the same patients. Moreover, upregulation of the autophagy-related genes in the diaphragm increases with the time spent on CMV.⁸

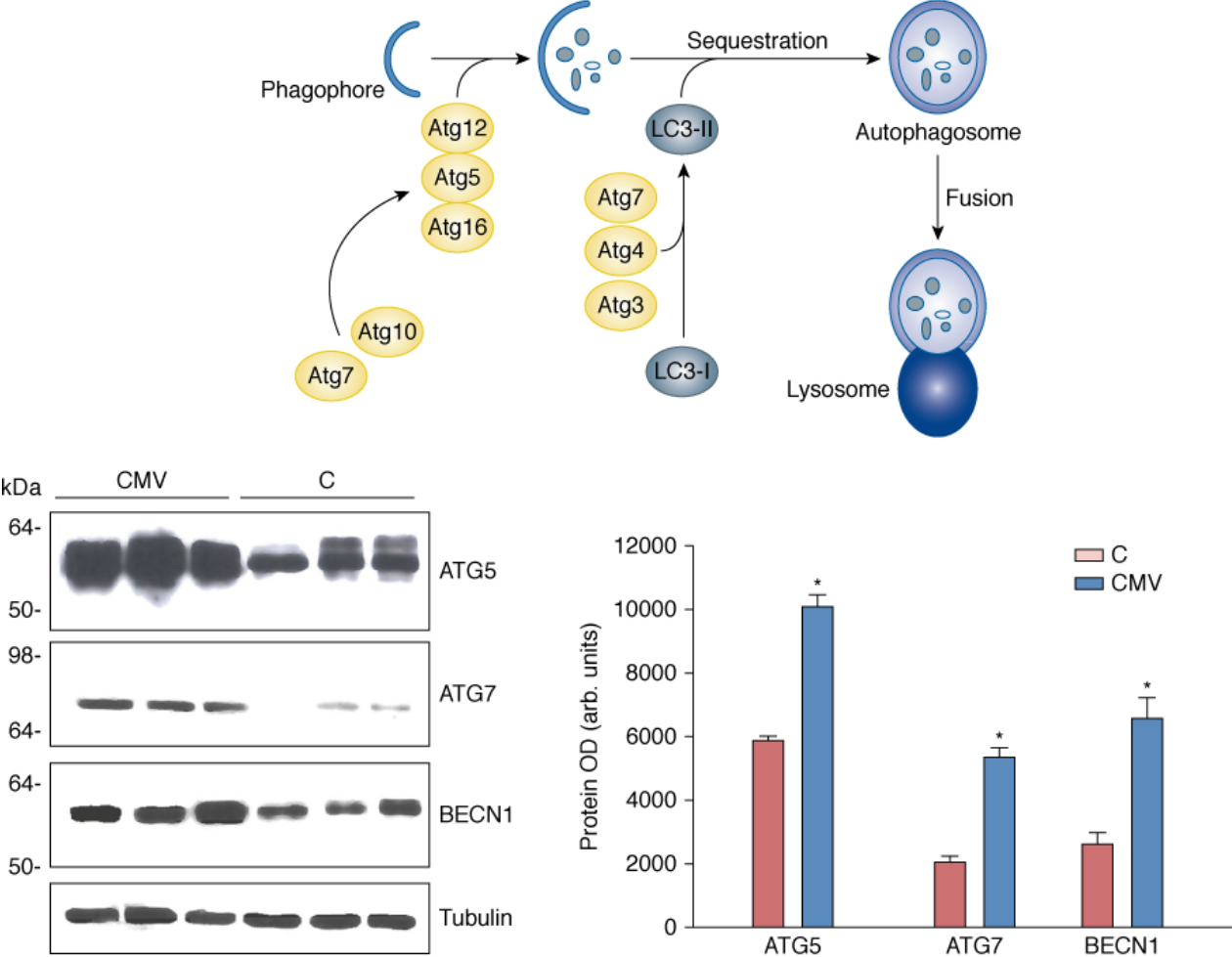
Figure 43-5



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Left panel. Representative electron micrograph of a section of a diaphragm from a control subject showing normal ultrastructure and absence of autophagosomes. Middle and right panels. Representative electron micrographs of a section of a diaphragm from a brain-dead organ donor undergoing controlled mechanical ventilation. These sections show autophagosomes (black arrows) in close proximity to mitochondria (M). G indicates glycogen particles. (Used, with permission, from Hussain et al.⁸)

Figure 43-6



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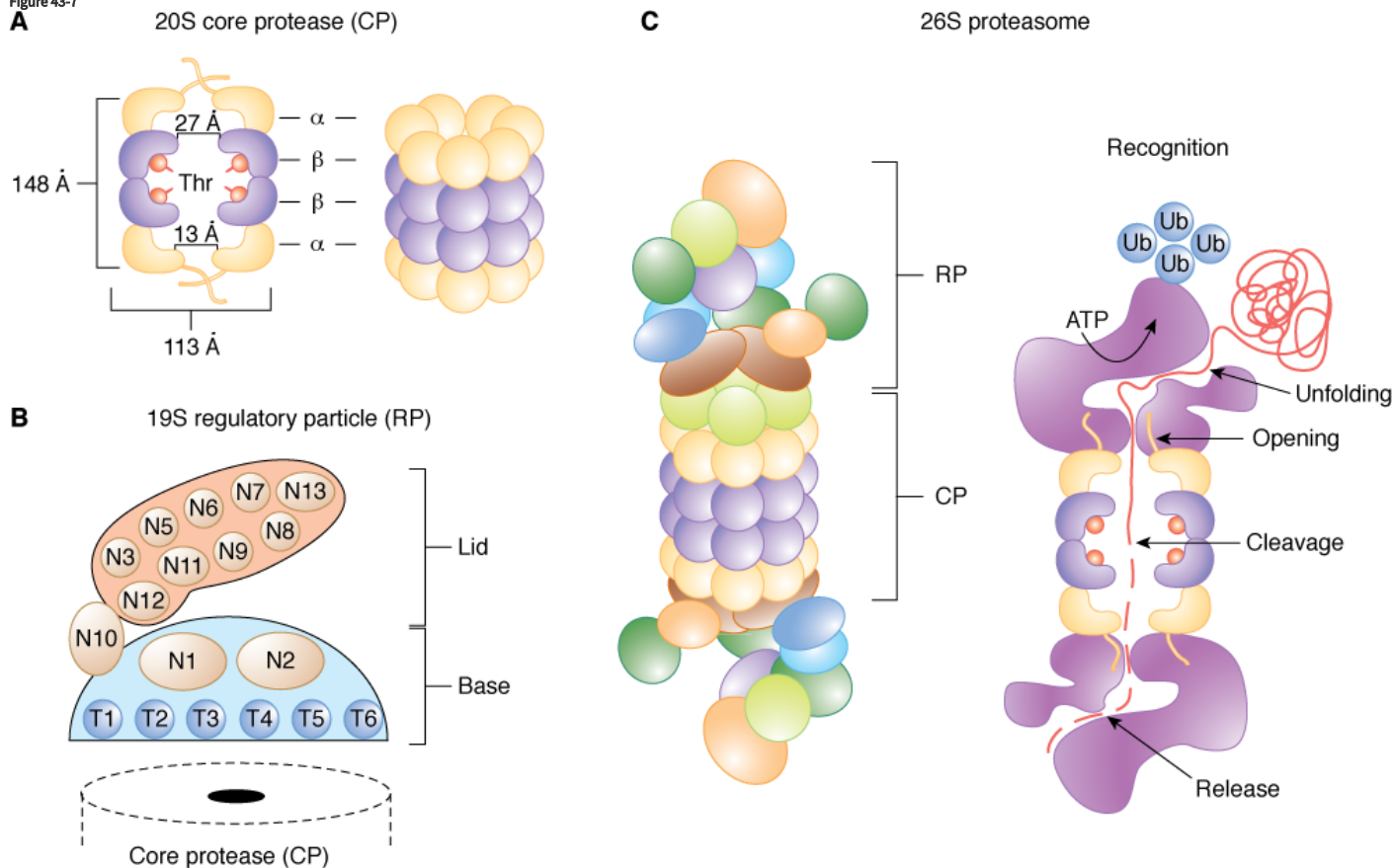
Autophagy-related genes in the human diaphragm after controlled mechanical ventilation (CMV). Top. Schematic cartoon of autophagosome assembly. Left. Representative immunoblots of BECN1, ATG5, and ATG7 in diaphragms of the control (C) and CMV groups. Right. Mean values of protein optical densities (OD) of BECN1, ATG5, and ATG7 in diaphragms of the C and CMV groups. *, $P < 0.05$ compared with control subjects; Atg, Autophagy related genes; LC3-I, LC3-II, microtubule-associated protein light chain-3 (LC3)-I or LC3-II; BECN1, beclin-1. (Used, with permission, from Hussain et al.⁸)

Proteasome

The proteasome is a multisubunit, multicatalytic complex that exists in two major forms (Fig. 43-7): The core 20S proteasome can be free or bound to a pair of 19S regulators to form the 26S proteasome, which degrades (in an adenosine triphosphate [ATP]-dependent manner) proteins covalently bound to a

polyubiquitin protein chain (ubiquitinated proteins). The binding of ubiquitin to protein substrates requires the ubiquitin-activating enzyme (E1), which utilizes ATP-derived energy to form a covalent link with a ubiquitin protein, followed by transfer of the active ubiquitin moiety to a ubiquitin-conjugating enzyme (E2), and finally transfer of this ubiquitin to the protein to be degraded via a ubiquitin ligase (E3) (Fig. 43-8). CMV increases the level of ubiquitin-protein conjugates in the diaphragm,^{6,25} which are the substrates of the 26S proteasome (Fig. 43-9). Key enzymes involved in the function of ubiquitin-proteasome pathway are upregulated in the human diaphragm after CMV, such as the skeletal muscle specific ubiquitin ligases (E3 enzymes), muscle atrophy F-box (MAFbx/Atrogin-1), and muscle ring finger-1 (MuRF1),^{5,8,25} as well as the E2 conjugases UBC2 and UBC4⁸ (Fig. 43-10). The changes in the E3 ligases are relatively specific to the diaphragm and were not observed in the quadriceps (control limb muscle) of the same patients, which were also inactive (Fig. 43-10).⁸

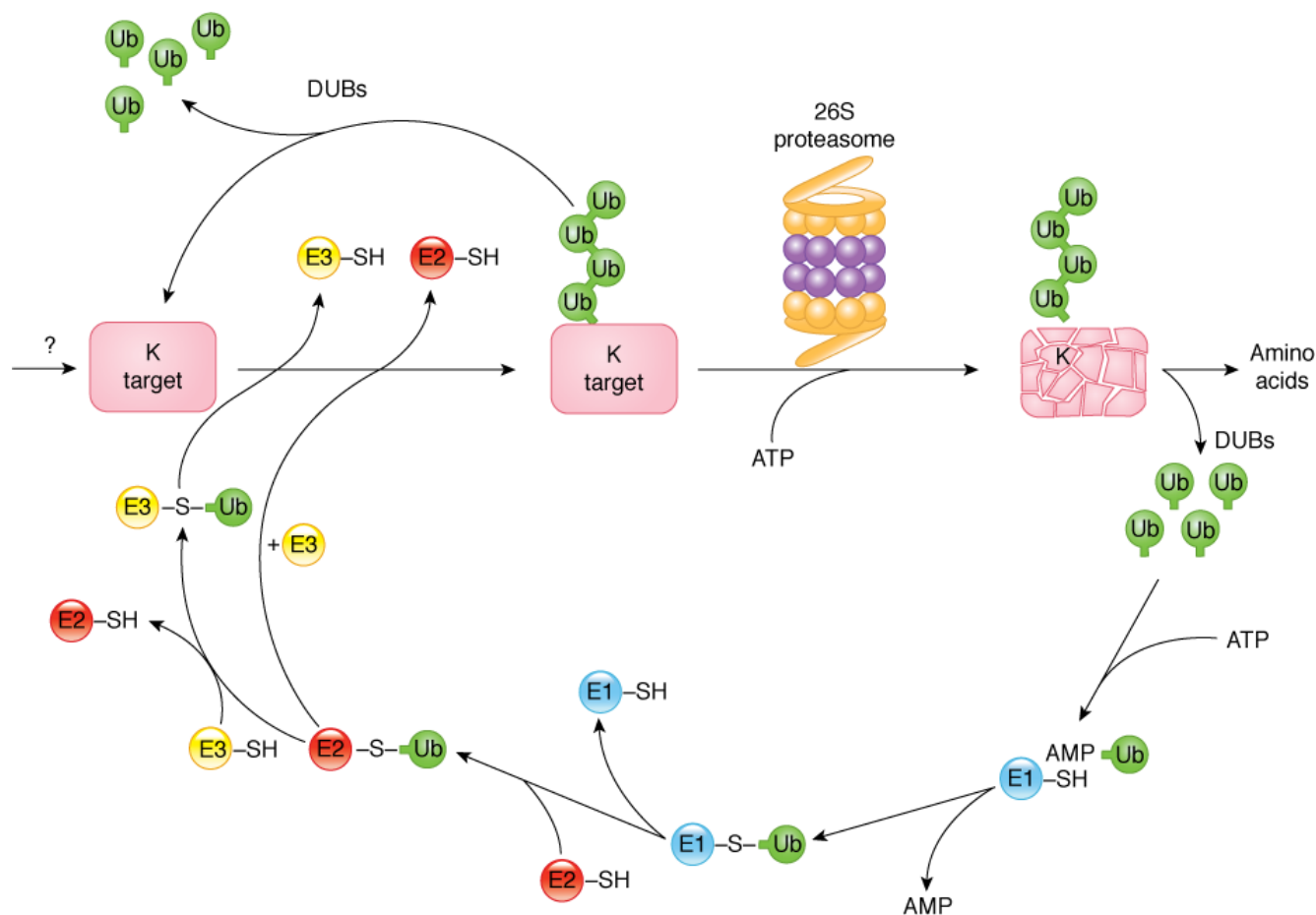
Figure 43-7



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The 20S and 26S proteasome isoforms. **A**. Organization of the core protease (CP) and **(B)** the regulatory particle (RP) with its *Lid* and *Base* subparticles. The dimensions of the CP were obtained from the crystal structure of the yeast complex. The *N*-terminal threonine residues that form the protease active sites in the β_1 , β_2 , and β_5 subunits are indicated. **C**. Proposed structure and sequence of events that lead to the degradation of a ubiquitinated protein by the 26S holoproteasome. *ATPase*, adenosine triphosphatase; *N*, RP non-ATPase subunits; *T*, RP AAA-ATPase subunits; *Thr*, threonine; *Ub*, ubiquitin. (Used, with permission, from Vierstra et al.⁷²)

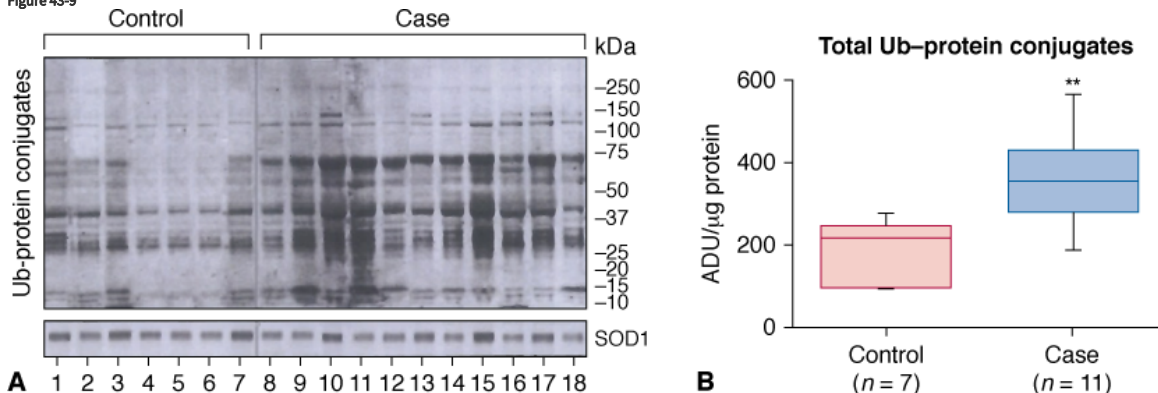
Figure 43-8



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The ubiquitin-proteasome pathway of proteolysis. Proteins degraded by the ubiquitin-proteasome pathway are first conjugated to ubiquitin (*Ub*). The process of linking ubiquitin to lysine residues in proteins destined for degradation involves the activation of ubiquitin by the E1 enzyme in an adenosine triphosphate (ATP)-dependent reaction. Activated ubiquitin is transferred to an E2 carrier protein and then to the substrate protein, a reaction catalyzed by an E3 enzyme (E3 ligase). This process is repeated as multiple ubiquitin molecules are added to form a ubiquitin chain. Once a conjugate is assembled bearing a chain of multiple Ubs, it is either recognized by the 26S proteasome and degraded in an ATP-dependent process or the conjugate is disassembled by deubiquitinating enzymes (*DUBs*), releasing the ubiquitin and target intact. In ATP-dependent reactions, ubiquitin-conjugated proteins are recognized and bound by the 19S complex, which releases the ubiquitin chain and catalyzes the entry of the protein into the 20S core proteasome. Degradation occurs in the core proteasome, which contains multiple proteolytic sites within its two central rings. The ubiquitin is not degraded but is released and reused. ADP, adenosine diphosphate; K, lysine; SH, sulfhydryl. (Used, with permission, from Vierstra et al.⁷²)

Figure 43-9

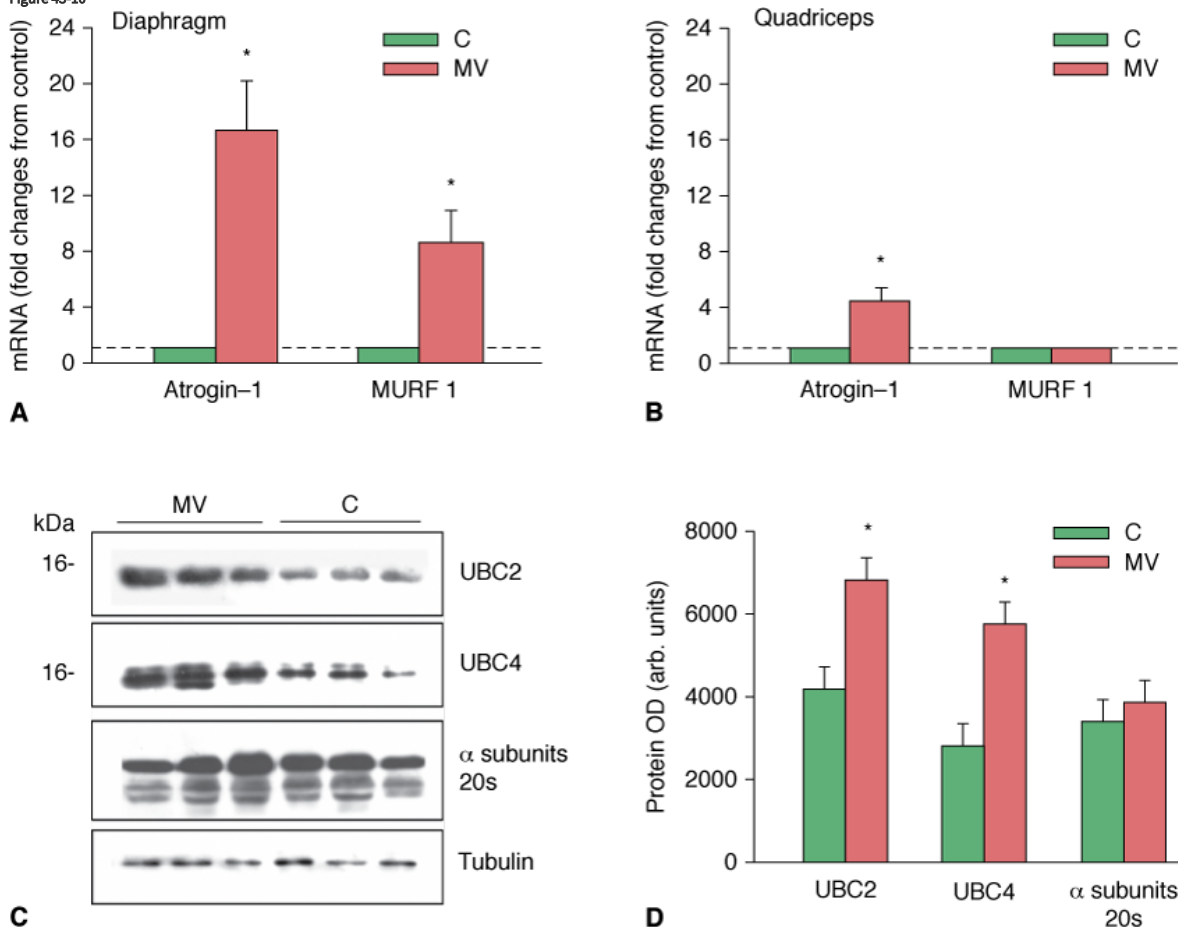


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Ubiquitin-protein conjugates in human diaphragmatic cytoplasm fractions after controlled mechanical ventilation (CMV). **A**, Immunoblotting results for ubiquitin (*Ub*)-protein conjugates in seven control and eleven case specimens. The fact that the control diaphragms show ubiquitination of many bands is consistent with the concept that ubiquitination is a normal process for protein turnover. The superoxide dismutase (*SOD*) 1 protein-loading control did not

differ between lanes. **B.** Case diaphragms exhibited approximately twice the amount of ubiquitinated proteins, suggesting that case diaphragms are exhibiting an upregulation of ubiquitination. **, $P < 0.01$; ADU, arbitrary density units. (Used, with permission, from Levine et al.²⁵)

Figure 43-10



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Proteasome ligase expression after controlled mechanical ventilation (CMV). **A.** Changes in messenger RNA (mRNA) expression levels of *Atrogin-1* and *MURF1* in the diaphragms of control (C) subjects and subjects undergoing CMV. *, $P < 0.05$ compared with control subjects. **B.** Changes in mRNA expression levels of *Atrogin-1* and *MURF1* in the quadriceps muscles of control subjects and subjects undergoing CMV. *, $P < 0.05$ compared with C subjects. **C.** Representative immunoblots of UBC2, UBC4, and α subunits of the 20S proteasome in diaphragms of the C and the CMV groups. **D.** Mean values of protein optical densities (OD) of UBC2, UBC4, and α subunits of the 20S proteasome in diaphragms of the C and CMV groups. *, $P < 0.05$ compared with control subjects. (Used, with permission, from Hussain et al.⁸)

Interestingly, the activity of the 20S proteasome is elevated in the human diaphragm after CMV.²⁵ The 20S proteasome is specialized in degrading proteins oxidized by reactive oxygen species. Oxidative damage of a protein results in its partial unfolding, exposing hidden hydrophobic residues.^{26,27} Therefore, an oxidized protein does not need to be further modified by ubiquitin conjugation to confer a hydrophobic patch, nor does it require energy from ATP hydrolysis to unfold. It should, however, be pointed out that the expression of various α subunits of the 20S proteasome was not increased in human diaphragms after CMV.⁸

Oxidative Stress

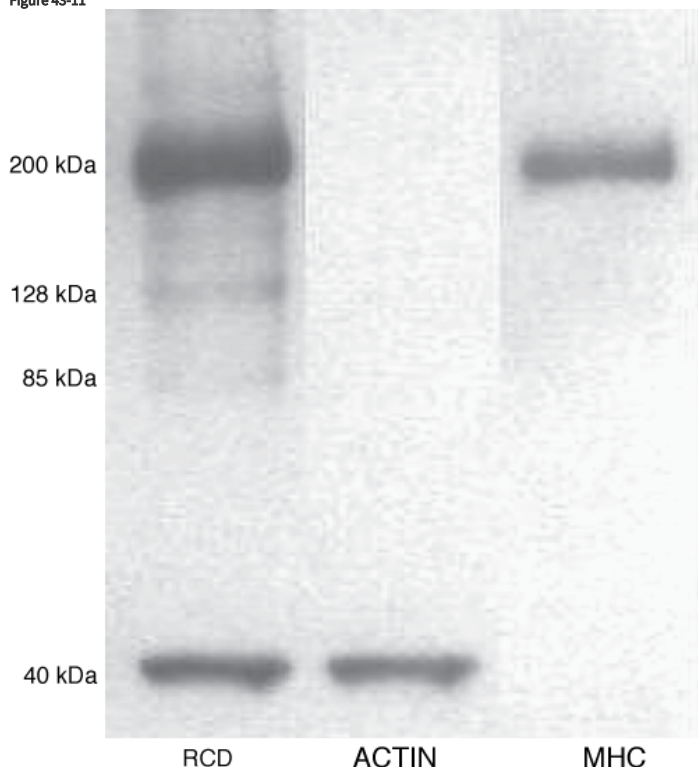
CMV is associated with augmented oxidative stress in the human diaphragm, as evidenced by the rise in protein oxidation (elevated 4-hydroxy-2-nonenal [HNE]-protein adduct formation⁸ and protein carbonylation,^{8,24} as well as the increased levels of superoxides documented by dihydroethidium staining.²⁴ Animal models of VIDD have also documented other indices of oxidative stress, such as lipid peroxidation (elevated 8-isoprostane,⁹ total lipid hydroperoxides,^{28,29} and thiobarbituric reactive substance content³⁰), and have provided direct evidence of oxidative stress by the increased emissions of dichlorofluorescein (a molecule that fluoresces upon reacting with reactive oxygen species within cells) when diaphragmatic strips from CMV-treated animals are incubated in vitro with the dye.³¹ Animal models of VIDD suggest that the onset of oxidative injury is rapid, occurring within 6 hours after the institution of CMV in rats²⁸ and is long lasting, being present after 3 days of CMV in piglets.³⁰ The mechanism(s) of oxidative stress generation remain elusive. Interestingly, the messenger RNA levels of both nuclear-encoded cytochrome C oxidase IV and the mitochondria-encoded cytochrome C oxidases I, II, and III

are reduced in the diaphragm after CMV. These observations indicate the presence of a functional impairment of the coupling between electron donors and acceptors as well as mitochondrial energy production,²⁴ and point to the mitochondria as potential sources of oxidative-stress generation in VIDD. The initial oxidative stress may result from the sudden change in the functional status of the diaphragm as it switches from continuous, intermittent contraction to passive rest (as occurs with CMV). Because noncontracting muscle fibers do not require major mitochondrial energy production, this altered functional status may place a “brake” on normal mitochondrial function. This may, in turn, cause mitochondrial reactive oxygen species accumulation and eventual leak.²⁴

The response of antioxidant enzymes in the diaphragm to CMV is controversial. Consistent with a mitochondrial role in this oxidative stress, the mitochondria-resident antioxidant gene, superoxide dismutase-2 (SOD 2), is significantly induced in the human diaphragm after CMV, whereas cytosolic superoxide dismutase-1 (SOD 1) is not.²⁴ Glutathione levels are reduced in the diaphragm, but this may simply be the result of oxidative-stress development.⁵

Oxidative stress can modify proteins involved in energetics, excitation–contraction coupling, intracellular calcium regulation, and force generation.³² In humans⁸ and in animal models of VIDD,²⁸ diaphragmatic protein oxidation was evident in proteins with molecular masses of about 40 and 200 kDa. These findings raise the possibility that actin (40 kDa) and/or myosin (200 kDa) undergo oxidative modification during CMV (Fig. 43-11), which is expected to compromise diaphragmatic contractility. This intriguing possibility awaits confirmation by more specific identification of the modified proteins. Interestingly, decreased levels of both myosin heavy chains and α -actin were documented in human diaphragms after CMV.²⁵

Figure 43-11



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Illustration of Western blots using monoclonal antibodies to identify oxidized proteins with molecular masses of approximately 200 to 40 kDa. *Left lane.* Reactive carbonyl derivatives (RCD), which are the footprints of protein modifications induced by oxidative stress in insoluble proteins isolated from the diaphragm of an animal exposed to controlled mechanical ventilation (CMV) for 18 hours. *Middle and Right lanes.* the same membrane stripped of the 2,4-dinitrophenylhydrazine antibody (the antibody recognizing RCD was removed) and then sequentially re-probed with monoclonal antibodies specific for rat skeletal muscle actin and all myosin heavy chain (MHC) isoforms. (Used, with permission, from Zergeroglu et al.²⁸)

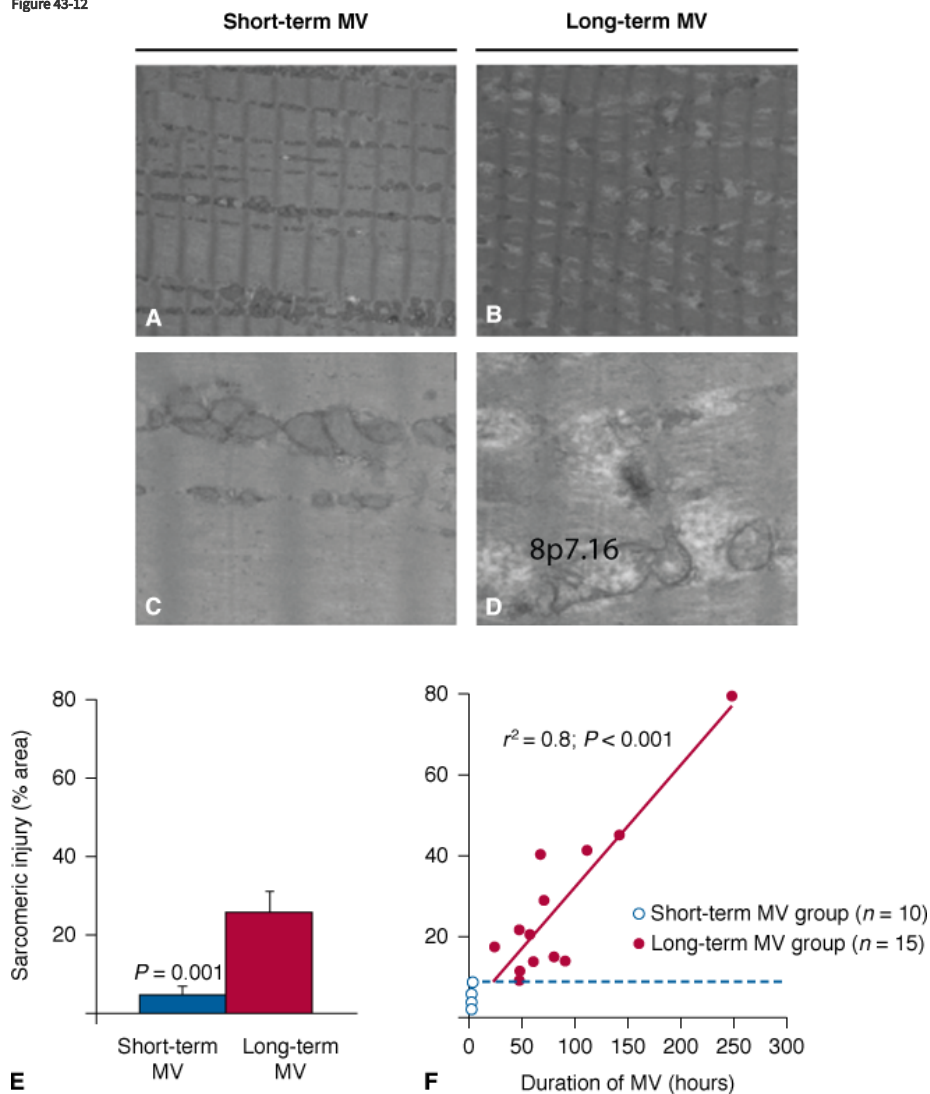
Oxidative stress, and especially 4-hydroxy-2-nonenal (produced in the diaphragm under conditions such as sepsis, resistive breathing,³³ and CMV⁸), can reduce the activity of plasma membrane calcium ATPase.³⁴ This would retard calcium removal from the diaphragmatic myofibers and contribute to calcium accumulation and calpain activation.¹⁸ Oxidative stress could injure various intracellular structures-organelles, activate proteolysis (especially through the 20S proteasome), and induce apoptosis.²⁴

Structural Injury

CMV leads to a significant increase in the prevalence of ultrastructural abnormalities in the human diaphragm, consisting of disruption of the normal myofibrillar organization with enlarged spaces containing disorganized sarcomeric material (Fig. 43-12). The longer the duration of CMV, the greater the

degree of injury in the human diaphragm.⁶ Animal models of VIDD have also documented structural abnormalities of different subcellular components of diaphragmatic myofibers, progressively developing after 2 to 3 days of CMV in rabbits.^{13,35,36} The changes consisted of disrupted myofibrils, increased numbers of lipid vacuoles in the sarcoplasm, and abnormally small mitochondria containing focal membrane disruptions. Similar alterations were observed in the external intercostal muscles of ventilated animals,³⁶ but not in hindlimb muscles.¹³ The structural abnormalities have detrimental effects on diaphragmatic contractility. The number of abnormal myofibrils is inversely related to the force output of the diaphragm.¹³ The mechanisms of injury have not been elucidated, but may involve activation of calpains, which have the ability to degrade several sarcomeric proteins, and direct cellular injury secondary to augmented oxidative stress.³⁷

Figure 43-12



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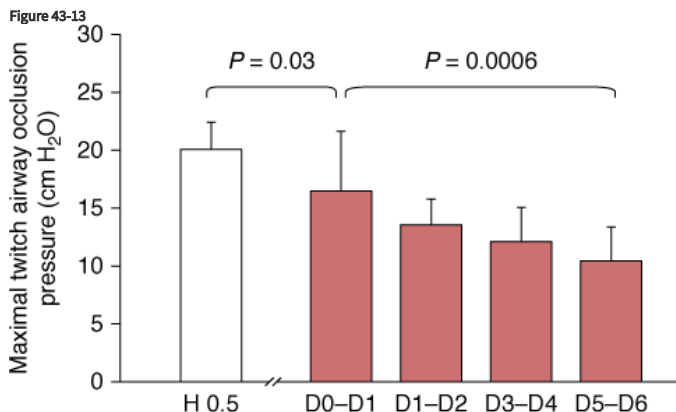
Relationship between duration of mechanical ventilation (MV) and diaphragmatic injury. Representative electron microscopy images of longitudinal ultrathin sections obtained from the diaphragms of the short-term MV (A and C) and long-term MV (B and D) groups are shown, at both low (A and B) and high (C and D) magnitude amplification. Note the disorganization of sarcomeric structure, which is only present in the long-term MV group images. E. Quantitative analysis of the prevalence of these findings in the two groups. F. Significant correlation between the degree of diaphragmatic injury and the duration of MV (horizontal dashed line indicates highest value of injury measured in short-term MV group). (Used, with permission, from Jaber et al.⁶)

Diaphragmatic Force and Endurance

Measurements in Humans

A major problem in documenting whether VIDD compromises diaphragmatic contractility in humans is the difficulty in accurately evaluating respiratory muscle function in critically ill patients undergoing mechanical ventilation, especially because these patients show variable levels of cooperation. An alternative, although technically demanding, approach is to obtain simultaneous recordings of esophageal (Pes) and gastric (Pga) pressures (their difference being transdiaphragmatic pressure [Pdi]), during bilateral magnetic phrenic nerve stimulation.^{38,39}

The magnitude of the negative deflection in tracheal pressure (Ptr) during twitch stimulation of the phrenic nerves against an occluded airway can be used as a surrogate of Pdi in intubated patients,^{39,40} and has the major advantage of not requiring the placement of esophageal and gastric balloon catheters. Jaber et al performed serial measurements of twitch Ptr in critically ill patients who were ventilated for approximately 1 week. Twitch Ptr decreased progressively during the period of CMV, with a mean reduction of $32 \pm 6\%$ at 6 days (Fig. 43-13).⁶ These data provide the best available evidence for the occurrence of contractile dysfunction as a component of VIDD in humans.



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Relationship between duration of mechanical ventilation (MV) and diaphragmatic function. Maximal twitch airway occlusion pressure generated by magnetic stimulation of the phrenic nerves at different time points in short-term MV (*open bar*; n = 6) and long-term MV (*solid bars*; n = 6) groups. D, number of days of MV; H, number of hours of MV. (Used, with permission, from Jaber et al.⁶)

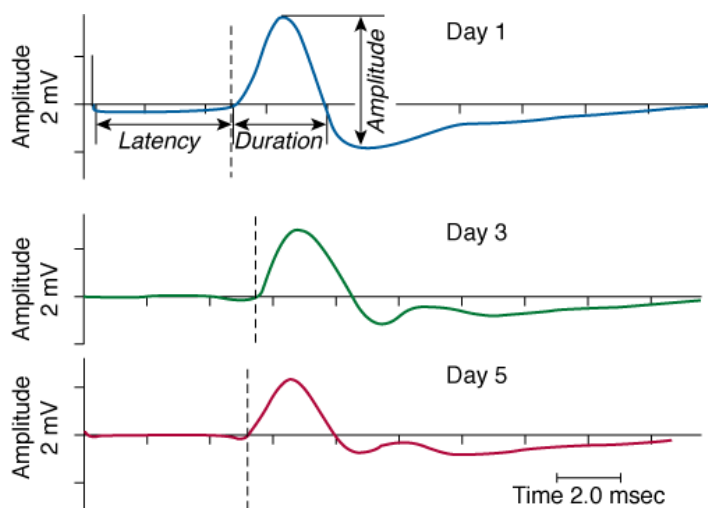
Several intriguing indirect data support this finding. Twitch Pdi elicited by magnetic stimulation of the phrenic nerves was reduced in mechanically ventilated patients³⁹ and in patients ready to undergo weaning trials,³⁸ as compared to normal subjects. Recently, Hermans et al⁴¹ assessed diaphragmatic force production in twenty-five mechanically ventilated intensive care unit patients using cervical magnetic stimulation of the phrenic nerves in a prospective single-center study. Seven patients were evaluated more than once during their intensive care unit stay. A longer duration of mechanical ventilation was associated with more severe diaphragmatic force loss, irrespective of the ventilator mode used. Impairment of diaphragmatic force was also correlated with the amount of sedation the patients received. The authors, however, were unable to determine whether sedative use is an independent risk factor for diaphragmatic weakness in ventilated patients or simply a marker of more prolonged periods of mechanical ventilation. It should be emphasized that the above observations are not specific evidence for the presence of VIDD, because other factors can lead to muscle weakness in the intensive care unit and modes other than CMV were used. Nevertheless, VIDD may have contributed to the reduction in diaphragmatic contractility in the patients who received full ventilator support (CMV).

Measurements in Intact Animals

CMV leads to decreased diaphragmatic force generating capacity in various animal species. In the intact diaphragm studied in vivo, Pdi generation upon phrenic nerve stimulation declines at all stimulation frequencies (20 to 100 Hz).^{13,30,42,43} The decline ensues early (1 day in rabbits,¹³ 3 days in piglets^{30,43}) and is progressive: Pdi decreasing to 63% of the control value after 1 day of CMV and to 49% of the control value after 3 days of CMV in rabbits.¹³ Within a few days (3 days in rabbits, 3 to 5 days in piglets, 11 days in baboons), the pressure-generating capacity of the diaphragm declines by 35% to 50%. The endurance of the diaphragm is also compromised.⁴²

The decrease in force-generating capacity that ensues with CMV is not caused by changes in lung volume³⁰ or abdominal compliance.^{42,43} Neural and neuromuscular transmission remain intact as evidenced by the lack of change in phrenic nerve conduction (latency) and stable response to repetitive stimulation of the phrenic nerve (Fig. 43-14).⁴³ In contrast, the compound muscle action potential declines progressively, suggesting that excitation-contraction coupling or membrane depolarization may be involved (Fig. 43-14).⁴³ Thus, the CMV-induced impairment in diaphragmatic force-generating capacity appears to reside within the myofibers.

Figure 43-14



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Neural and neuromuscular transmission to the diaphragm during experimental VIDD. The evoked compound muscle action potential (CMAP) tracings of the diaphragm upon electrical stimulation of the phrenic nerves in vivo from one piglet on days 1, 3, and 5 of CMV, respectively. The time from the stimulus to the onset of CMAP (latency) does not change after 3 to 5 days of CMV, whereas the amplitude of CMAP is progressively reduced. This indicates that the neural and neuromuscular transmission are not affected when VIDD develops and that the contractile dysfunction resides within the diaphragmatic myofibers. (Used, with permission, from Radell et al.⁴³)

In Vitro Measurements

The isometric (both twitch and tetanic) tension development by isolated diaphragmatic strips in vitro^{10–12,43,44} confirm the in vivo findings and suggest that the decline in contractility is an early (12 hours) and progressive phenomenon,⁴⁴ the isometric force declining by 30% to 50% after 1 to 3 days of CMV in rats. The effects of CMV on diaphragm in vitro fatigability are controversial.

Drugs and Ventilator-Induced Diaphragm Dysfunction

All information regarding the role of drugs in the development of VIDD comes from animal models. Anesthetics can be excluded as causes of VIDD because studies that used appropriate controls (to the extent feasible, viz., a group of anesthetized spontaneously breathing animals) concluded that the decreased contractility was the result of mechanical ventilation per se (and not anesthetic use).¹³

Neuromuscular blockers^{11,42} cannot solely account for the decreased contractility secondary to CMV, because decreased contractility was also observed in studies that did not use these drugs.^{13,44} Nevertheless, the effects of 24 hours of CMV and aminosteroidal neuromuscular blockers (rocuronium) are synergistic in depressing diaphragmatic contractility, in inducing atrophy of type IIx/b fibers, and in upregulating the ubiquitin ligase (E3) muscle ring finger-1 (MuRF1) (but not the E3 muscle atrophy F-box/Atrogin-1)²⁹—a synergism not observed with different doses of benzylisoquinoline neuromuscular blockers (cisatracurium).⁴⁵

Clinical Relevance

Clinical Context

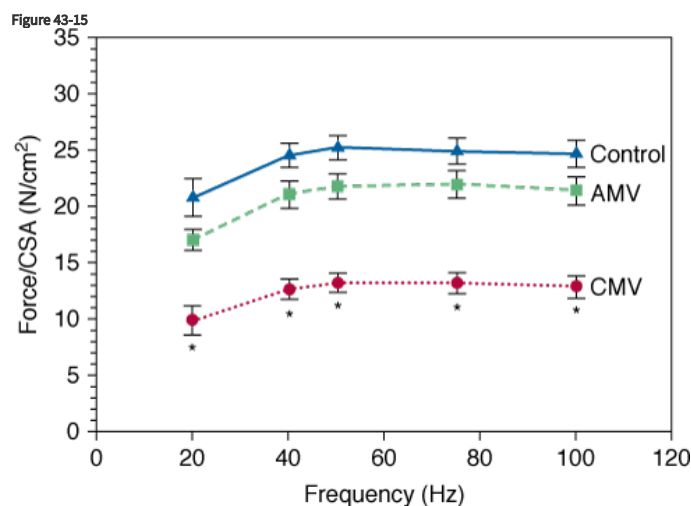
VIDD should be suspected in patients who fail to wean after a period of CMV. The weaning failure is related to respiratory muscle weakness. Other causes of respiratory muscle weakness should be ruled out,⁴⁶ although several may coexist with VIDD.

Ventilator-Induced Diaphragm Dysfunction Prevention

Ventilator Strategy

Because data in humans are lacking, suggestions are based on animal models and speculations. The time spent in CMV must be curtailed to the extent possible, especially in older individuals, because the effects of aging and CMV are additive.⁴⁷ Although CMV induced similar losses (24%) in diaphragmatic isometric tension in both young and old animals, the combined effects of aging and CMV resulted in a 34% decrement in diaphragmatic isometric tension as compared to young control animals.

When feasible, partial support modes should be used. Partial support modes can be used in conditions traditionally considered as indications for CMV, such as acute lung injury and/or acute respiratory distress syndrome^{48,49} although complete CMV with the use of neuromuscular blockers in the first 48 hours of severe acute respiratory distress syndrome may be beneficial.³ In a rabbit animal model, assisted (flow-triggered pressure limited) ventilation from the onset of ventilator support resulted in attenuation of the force loss induced by CMV (Fig. 43-15).⁵⁰ Similarly, adaptive support ventilation was able to prevent any deleterious effect of mechanical ventilation on the piglet diaphragm.⁵¹ Thus, it stands to reason that preserving diaphragmatic contractions during mechanical ventilation should attenuate the force loss induced by CMV, although other forms of partial ventilator support (pressure support, synchronized intermittent mandatory ventilation) have not been experimentally tested. Pressure support was able to prevent increased proteolysis in the diaphragm when compared to CMV, although its effects on contractility were not reported.⁵² It should be stressed that these suggestions originate from studies in healthy animals. The extent to which these same phenomena occur in critically ill patients with various diseases is not known. During sepsis in rats, albeit of short duration (4 hours), CMV protected the diaphragm against injury,⁵³ which raises the question about the effects of CMV on the diaphragm of septic (not healthy) humans.



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Diaphragmatic tetanic force at various stimulation frequencies in control, assisted mechanical ventilation (AMV), and controlled mechanical ventilation (CMV) in rabbits. Values are mean $\pm$ SE (standard error). *, $p < 0.01$, CMV versus control and AMV; CSA, cross-sectional area. (Used, with permission, from Sassoon et al.⁵⁰)

Assisted modes or even noninvasive ventilation in hypercapnic patients with chronic obstructive pulmonary disease^{54–56} is an alternative to the use of CMV in weaning-failure patients⁵⁷—a strategy based on the premise that respiratory muscle fatigue (requiring rest to recover) is the cause of weaning failure.^{58,59} Fatigue is expected to occur because the load that the respiratory muscles of weaning-failure patients is increased to the level that would predictably produce fatigue of the respiratory muscles⁶⁰ if patients were allowed to continue spontaneous breathing without ventilator assistance. Evidence, however, does not support the existence of low-frequency fatigue (the type of fatigue that is long-lasting, taking more than 24 hours to recover) in patients who fail to wean despite the excessive respiratory muscle load.³⁸ This is because physicians typically terminate a weaning trial and place a patient back on the ventilator before low-frequency fatigue has had time to develop. Thus, no reason exists to completely unload the respiratory muscles with CMV for fatigue reversal when weaning is typically terminated.⁵⁷

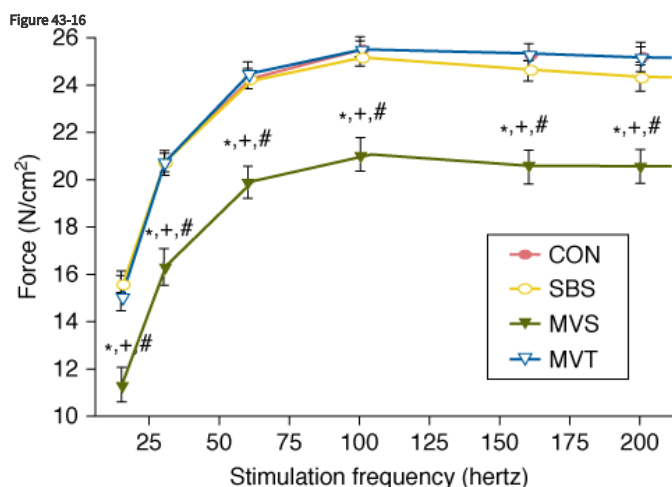
Intermittent Diaphragmatic Contractions

When CMV is inevitable, short periods of diaphragmatic activity have been suggested as preventive countermeasure. This could be achieved with either phrenic nerve stimulation or short periods of intermittent spontaneous breathing. As little as 30 minutes of pacing of one hemidiaphragm each day attenuated atrophy in this hemidiaphragm during prolonged CMV in a tetraplegic patient compared to the nonpaced hemidiaphragm.⁶¹ In rats subjected to 24 hours of CMV, either 5 or 60 minutes of spontaneous breathing every 6 hours did not preserve diaphragmatic force. Rats receiving CMV developed reduced cross-sectional areas of type I and type IIx/b diaphragmatic fibers, which was not observed in intermittently spontaneously breathing rats, yet no difference was observed in the cross-sectional areas between the CMV and intermittently spontaneously breathing rats.⁶² Whether more frequent or longer intervals of spontaneous breathing might be more effective in preventing VIDD awaits experimental proof.

Pharmacologic Approaches

Antioxidant supplementation could decrease oxidative stress and thus could attenuate VIDD. Accordingly, when rats were administered the antioxidant Trolox (6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid, a water-soluble analog of vitamin E) (Fig. 43-16) or apocynin (a nicotinamide adenine dinucleotide

phosphate oxidase inhibitor with antioxidant properties) from the onset of CMV, its detrimental effects on contractility and proteolysis were prevented.^{63,64}



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Force-frequency curves of in vitro diaphragm strips from control (CON), spontaneously breathing (SBS), mechanical ventilation (MVS), and mechanical ventilation animals receiving Trolox (MVT). Values represent means $\pm$ SEM (standard error of mean). *, Significantly different from CON group, $p < 0.05$; +, significantly different from SBS group, $p < 0.05$; #, significantly different from MVT group, $p < 0.05$. (Used, with permission, from Betters et al.⁶³)

A similar approach is adopted by nature itself! Various dormant animals immobilized for prolonged periods of time prevent muscle atrophy through a decrease in metabolic rate that reduces reactive oxygen species formation, and a concomitant rise in antioxidant enzymes.^{65,66} Interestingly, a combination of vitamins E and C administered to critically ill surgical (mostly trauma) patients was effective in reducing the duration of mechanical ventilation as compared to nonsupplemented patients.⁶⁷ It is tempting to speculate that part of this beneficial effect was mediated by preventing VIDD. Thus, when CMV is used, concurrent administration of antioxidants seems justified and a meta-analysis suggests that they are beneficial in critical care patients.⁶⁸

Administration of leupeptin (an inhibitor of lysosomal proteases and calpain) at the beginning of CMV prevented the development of diaphragmatic contractile dysfunction and atrophy²⁰ in experimental animals. This raises the possibility of future clinical trials of protease inhibitors in patients to prevent VIDD.

Recovery from Ventilator-Induced Diaphragm Dysfunction

There is no established or experimentally tested therapy for VIDD. Theoretically, resumption of spontaneous breathing would retrain the respiratory muscles, yet the time course of recovery of normal function is unknown. A major concern is that diaphragmatic disuse associated with CMV would increase its susceptibility to subsequent contraction-induced injury once respiratory efforts are resumed, similar to other skeletal muscles.⁶⁹ Rats receiving 24 hours of CMV exhibited a 26% decline in maximal specific diaphragmatic force with no apparent injury to the cell membrane or evidence of inflammation.⁷⁰ Resumption of spontaneous breathing for 2 hours did not exacerbate contractile dysfunction in these rats or induce membrane injury or macrophage invasion.⁷⁰ Reloading, however, was associated with increased myeloperoxidase activity and neutrophil infiltration in the diaphragm, which is expected to cause injury at a later time point, if reloading were continued.⁷⁰ Further studies are needed to elucidate the recovery response of the diaphragm that has developed VIDD as spontaneous respiratory muscle activity is resumed.

Important Unknowns

All data on animal models of VIDD have been obtained in previously healthy animals. Human data also have been obtained in brain-dead organ donors who did not have severe infections. Thus, the extent to which the deleterious effects of CMV on diaphragmatic function and biology occur in critically ill patients is unknown.

Modes of partial ventilation support attenuate the deleterious effects of CMV in animal models of VIDD. Whether this is also the case in humans is a clinically important question. Furthermore, the actual amount of assist necessary to prevent VIDD is also unknown.

Pharmacologic approaches used successfully to prevent VIDD in animal models have not been tested in human trials. Thus, no drugs has demonstrated clinical efficacy in the prevention or treatment of human VIDD.

Once VIDD is suspected, the best approach is to reload the diaphragm with abrupt or gradual reduction in the level of ventilator assistance, although the optimal rate of assist reduction is not known.

The Future

Microarray analysis has identified 354 differentially expressed gene products in the diaphragms of animals subjected to CMV compared to control animals.⁷¹ Intense research is required to unravel the mechanisms of VIDD. Animal models can be used to study the effects of CMV on the diaphragm of a critically ill organism, and to determine the best approach to reload the diaphragm once VIDD has developed. Clinical trials of different ventilator strategies (different levels of assist) and of pharmacotherapy to prevent VIDD should be designed and conducted. Finally, pharmacologic approaches for VIDD recovery are the next logical step of research in the field.

Summary and Conclusion

Over the last two decades researchers have discovered that mechanical ventilation can damage the previously normal respiratory muscles. CMV imposes a unique form of skeletal muscle disuse: The diaphragm is simultaneously unloaded, electrically quiescent, and phasically shortened by cyclical lung inflation or tonically shortened when PEEP is used. The respiratory muscles are not some inert mechanical pump that can be replaced cavalierly by the ventilator. The respiratory muscles should remain as active as possible because they are plastic and vulnerable.

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